Lab: Experiments and Evidence -- Data Investigation

Objective

> **Learning Goal:** Use data collection and analysis to explore the concept of testing hypotheses through probability and pattern-finding.

This is a **guided worksheet** designed for middle school students (Grades 6-8).

Materials Needed

- Pencil and worksheet
- Calculator (optional)
- Graph paper or blank paper
- Dice, coins, or random number generator (optional)
- Lab journal or notebook

Part 1: Data Collection (10 min)

Design and run a simple classroom experiment. Record your data carefully -- every number matters.

Data Point #	What you measured/observed	Category or value	Notes
1			
2			
3			
4			
5			
6			
7			
8			

Write your response here

Part 2: Pattern Analysis (15 min)

Look at your data and search for patterns. A math detective does not just collect numbers -- they ask what the numbers MEAN.

Analysis Question	Your Finding
What is the most common value or category?	
What is the range (highest - lowest)?	
Do you see any patterns or trends?	
Are there any surprising outliers?	
What would you predict the next data point would be?	

Write your response here

Part 3: Prediction Challenge (10 min)

Based on your data, make a prediction and then test it.

Step	Your Work
My prediction	
My confidence level (1-10)	
How I tested it	
What actually happened	
Was my prediction correct?	
How would I update my prediction now?	

Write your response here

Part 4: Reflection Table

Question	Your Answer
What math skills did you use in this investigation?	
How does data help you see patterns you would miss otherwise?	
What is the connection between data and prediction?	
How is this related to Active Inference?	
What would you investigate next?	

Write your response here